

Uptime.c

```
zmerchant7@DESKTOP-3KASI x + v
iltin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/uptime.o user/uptime.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_uptime user/uptime.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_uptime > user/uptime.asm
riscv64-linux-gnu-objdump -t user/_uptime | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/uptime.sym
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mmodel=medany -ffreestanding -fno-common -nostdli
b -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-b
zero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-bu
iltin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/history.o user/history.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_history user/history.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
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mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_regex user/_init user/_kill user/_ln user/_ls
user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user
/_sixfive user/_memdump user/_find user/_uptime user/_history user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1234 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=no
ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
33 ticks
$
```

History.c

```
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iltin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/uptime.o user/uptime.c
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user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep user
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ne,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
33 ticks
$ history
uptime
history
$
```

Find . grep.c

A screenshot of a Windows 10 desktop. The background is a dark blue gradient. In the foreground, a black terminal window is open, displaying a series of commands and their outputs. The commands include 'mkdir a', 'echo a > foo', 'echo a> fcobar', 'echo a> bar', 'find . -E bar', and 'find . -E foo'. The terminal output shows an error for 'mkdir: a failed to create' and successful execution for the other commands. A notification bubble from the 'Snipping Tool' is visible in the bottom right corner, stating 'Screenshot copied to clipboard' and 'Automatically saved to screenshots folder.' The taskbar at the bottom shows various application icons, including the Start button, Search, and several open applications. The system tray in the bottom right corner displays the date and time as '12:13 AM 9/27/2023'.

Make Grade

```
zmerchant7@DESKTOP-3KASLBT: ~$
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (5.2s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (0.7s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.0s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (1.0s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (3.5s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.1s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (0.6s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (0.8s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.0s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.2s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.8s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.0s)
== Test exec, recursive find ==
$ make qemu-gdb
Timeout! exec, recursive find: OK (33.3s)
== Test time ==
time: OK
Score: 131/131
zmerchant7@DESKTOP-3KASLBT: ~/xv6-labs-2025$
```

WAIT AND TAB TEST WORKS (NO SS)